

Automatic Infrared FAUCET CONTROLLER

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Described here is an easy-to-use circuit of an automatic infrared (IR) faucet controller suitable for hand hygiene and water conservation at homes, hospitals and offices. With this compact controller you can turn on and off a faucet automatically, preventing water wastage and saving energy costs.

The core part of the faucet controller is an electronic circuitry that determines the opening of the valve. An IR sensor continuously monitors objects like hands in front of the faucet. When you place your hand in front of the faucet, it automatically activates an electromagnetic relay to switch a solenoid valve, and concerted action provides enough water.

Another advantage of this battery-operated faucet controller is that, it

can be installed on an existing water tap. The author's prototype on a breadboard is shown in Fig. 1.

Circuit and working

The circuit diagram of the automatic IR faucet controller is shown in Fig. 2. It is built around Schottky diode 1N5817 (D1), optical sensor (OS1), pnp transistor S8550 (T1), two npn S8050 transistors (T2, T3), rectifier diode 1N4007 (D2), 5V-1CO relay, 6V solenoid valve and a few other components.

The circuit is constructed using a reflective optical sensor TCRT5000. It includes a 950nm IR emitter and phototransistor in a leaded package (with daylight blocking filter), which blocks visible light to a certain extent.

It is powered by 4.8V, drawn from a series of four nickel-metal-hydride (Ni-MH) rechargeable cells (1.2V each). External 6V DC power supply connected across connector CON1 is also provided as an alternate for running the circuit.

Power supply is protected by 1N5817 (D1). 6V can be used to recharge the battery pack, with only minor modifications. Switch S1 is used for turning the circuit on and off.

Resistor R1 controls the operating current of the IR inside TCRT5000 (OS1) within safe limits. R2 deter-

mines the detection sensitivity of the IR sensor. The IR emits an invisible constant beam of IR light. When a hand or object crosses this beam, some IR light is reflected back to the sensor.

This reflection is detected by the IR receiver (inbuilt phototransistor), which results in a voltage drop across R2. This switches off S8050 (T2). Next, S8550 (T1) and S8050 (T3) support efficient driving of the 5V electromagnetic relay (RL1) when a valid proximity detection is made.

The electromagnetic relay enables and disables the flow of water through the connected solenoid valve. The 10µF capacitor (C1) ensures a turn-off delay of about six seconds, so that the circuit does not stop water flow as soon as the proximity sensor is turned off.

LED1 works as active-state indicator. 1N4007 (D2) adds effective circuit protection, as it provides a path for the dissipation of stored energy (without flowing back) when RL1 is de-energised.

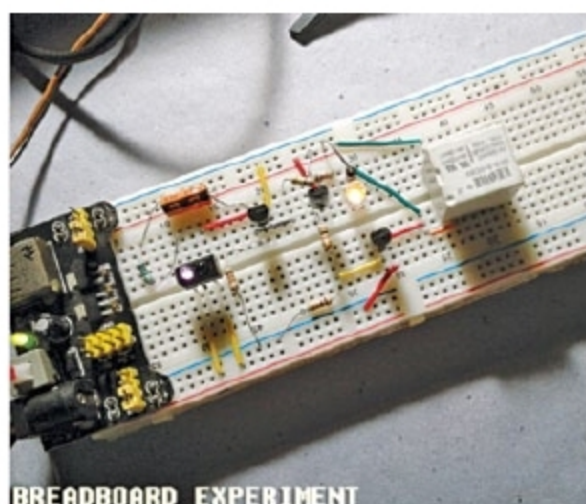


Fig. 1: Author's prototype

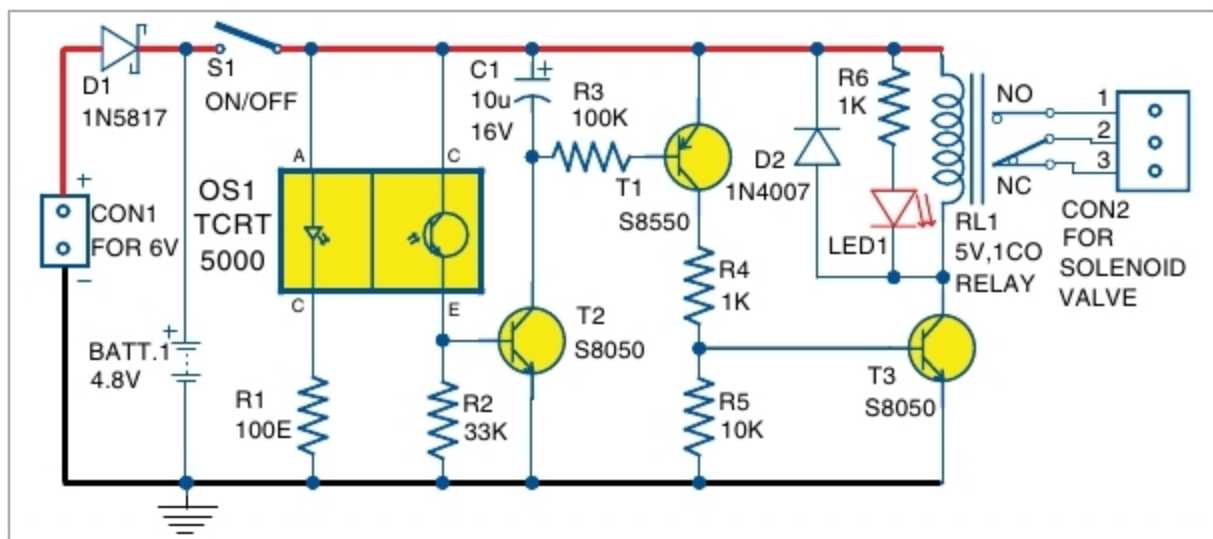


Fig. 2: Circuit diagram of automatic IR faucet controller

PARTS LIST

Semiconductors:

OS1	- TCRT5000 reflective optical sensor
T1	- S8550 pnp transistor
T2, T3	- S8050 npn transistor
D1	- 1N5817 Schottky diode
D2	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1	- 100-ohm
R2	- 33-kilo-ohm
R3	- 100-kilo-ohm
R4, R6	- 1-kilo-ohm
R5	- 10-kilo-ohm

Capacitors:

C1	- 10µF, 16V electrolytic
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Miscellaneous:

CON1	- 2-pin terminal connector
CON2	- 3-pin connector
BATT.1	- 4.8V battery pack/ 1.2V×4 Ni-MH cell
S1	- On/off switch
RL1	- 5V, 1CO relay - 6V solenoid valve - 6V regulated power supply/ battery